In May 2005, Kittilä and Ivalo flooded together.

The snow had warned us two months earlier.

The 1 March snow flagged 2005 in advance. Then both rivers flooded in the same spring.



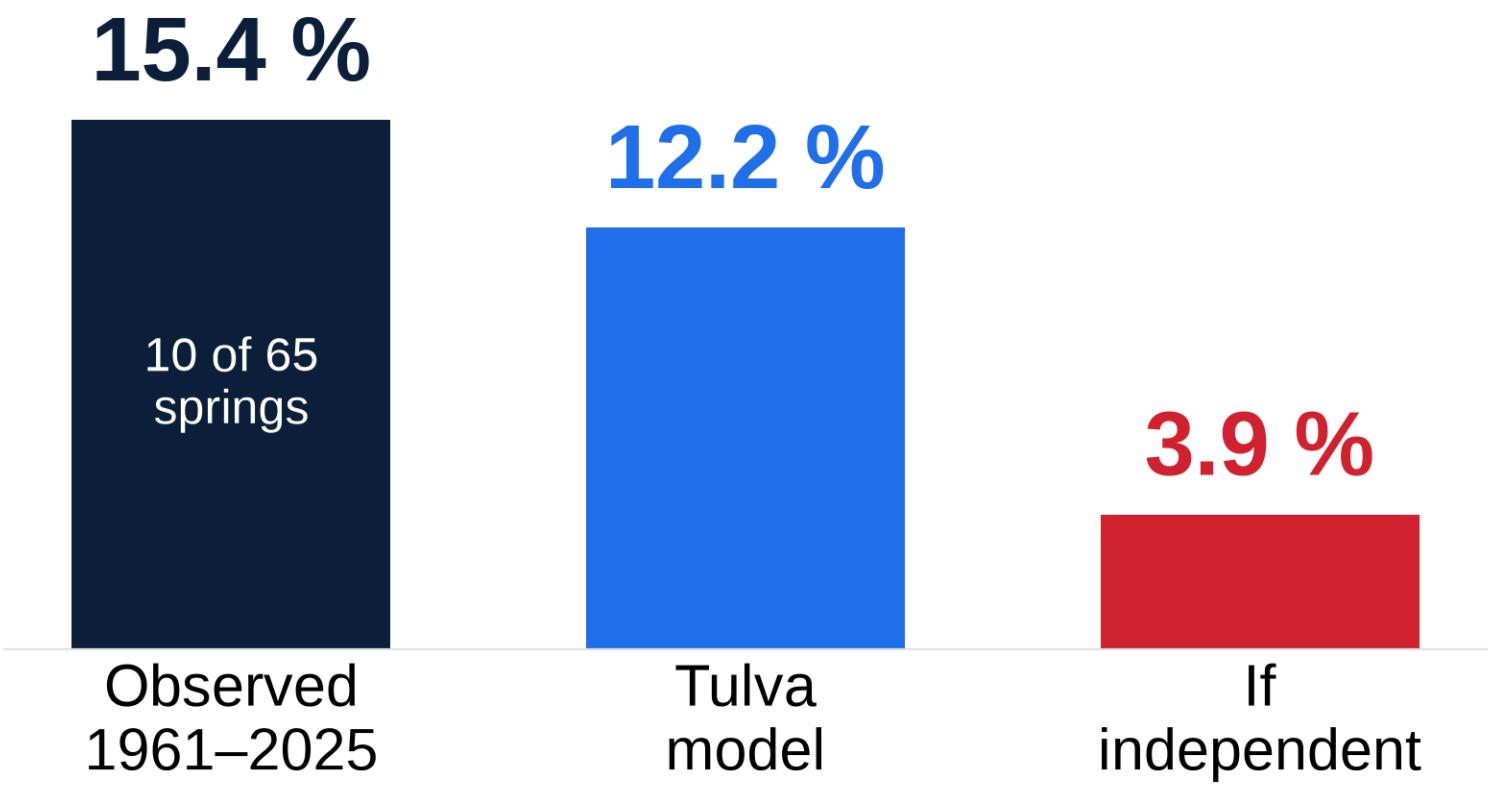
Tulva app: spring 2005 replay

Source: SYKE discharge 1961–2025 and 1 March snow (SWE); Tulva back-test (out/backtest.json)

THE GAP

Finland's rivers flood together, not one at a time

How often do ≥4 of 14 river areas flood in the same spring?



Springs with ≥4 of 14 areas over their 1-in-10 level

15.4 %
observed (10 of 65 years)

12.2 %
Tulva joint model

3.9 %
if areas were independent

Reinsurance renews 1 January, before the snow signal arrives.

Source: SYKE 1961–2025 (+ SMHI Torne), Tulva model (exact); vesi.fi / OP (1-in-50 rule)

THE PRODUCT

Tulva Index: bought in winter, pays in spring

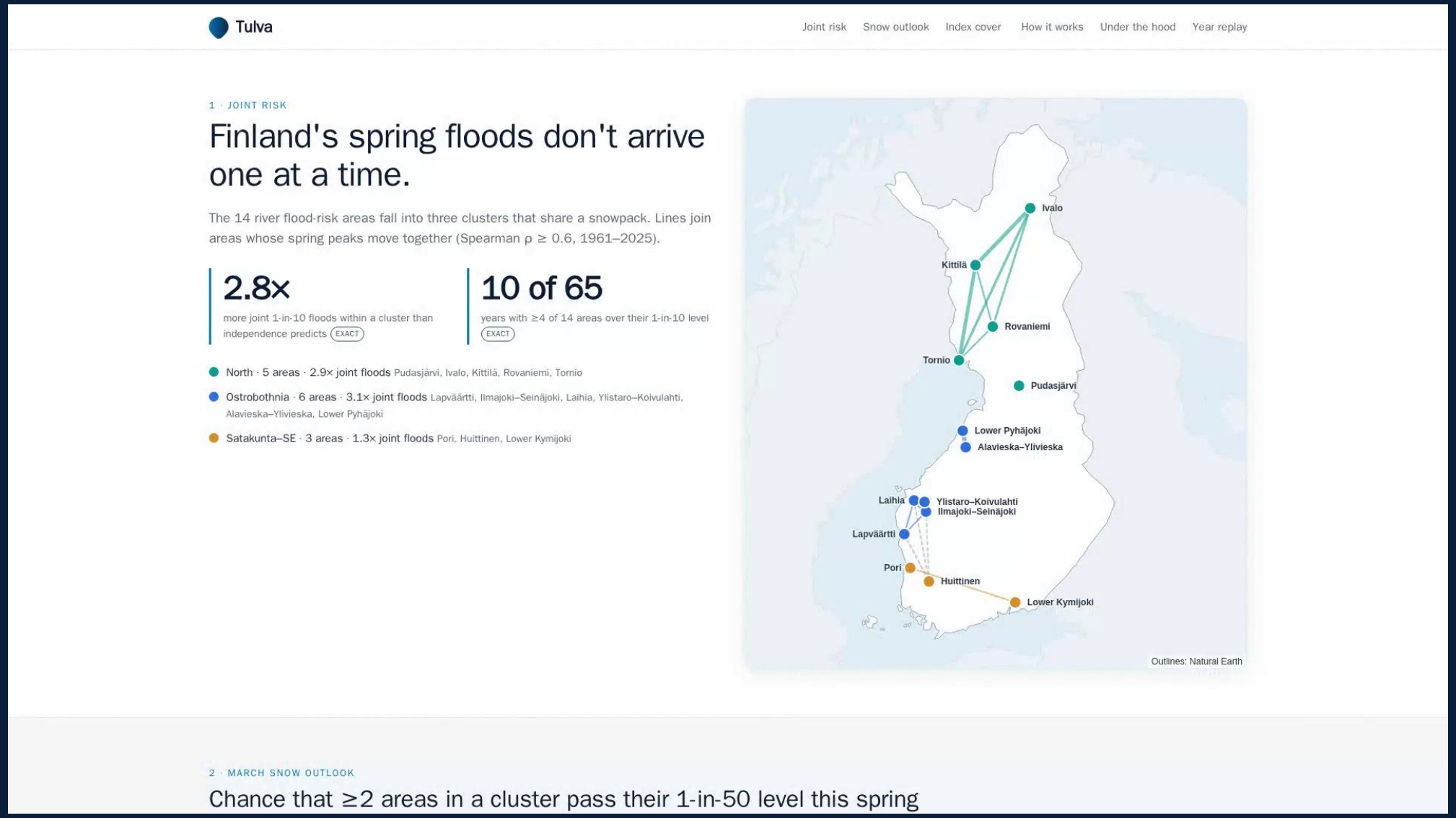
Pays €10M if ≥2 of 5 northern gauges pass their 1-in-50 level this spring

MARKET PRICE PER YEAR



Today's cat models assume how areas flood together. **Tulva measures it from 65 years of river data, ties it to this winter's snow, and prices it as a cover.**

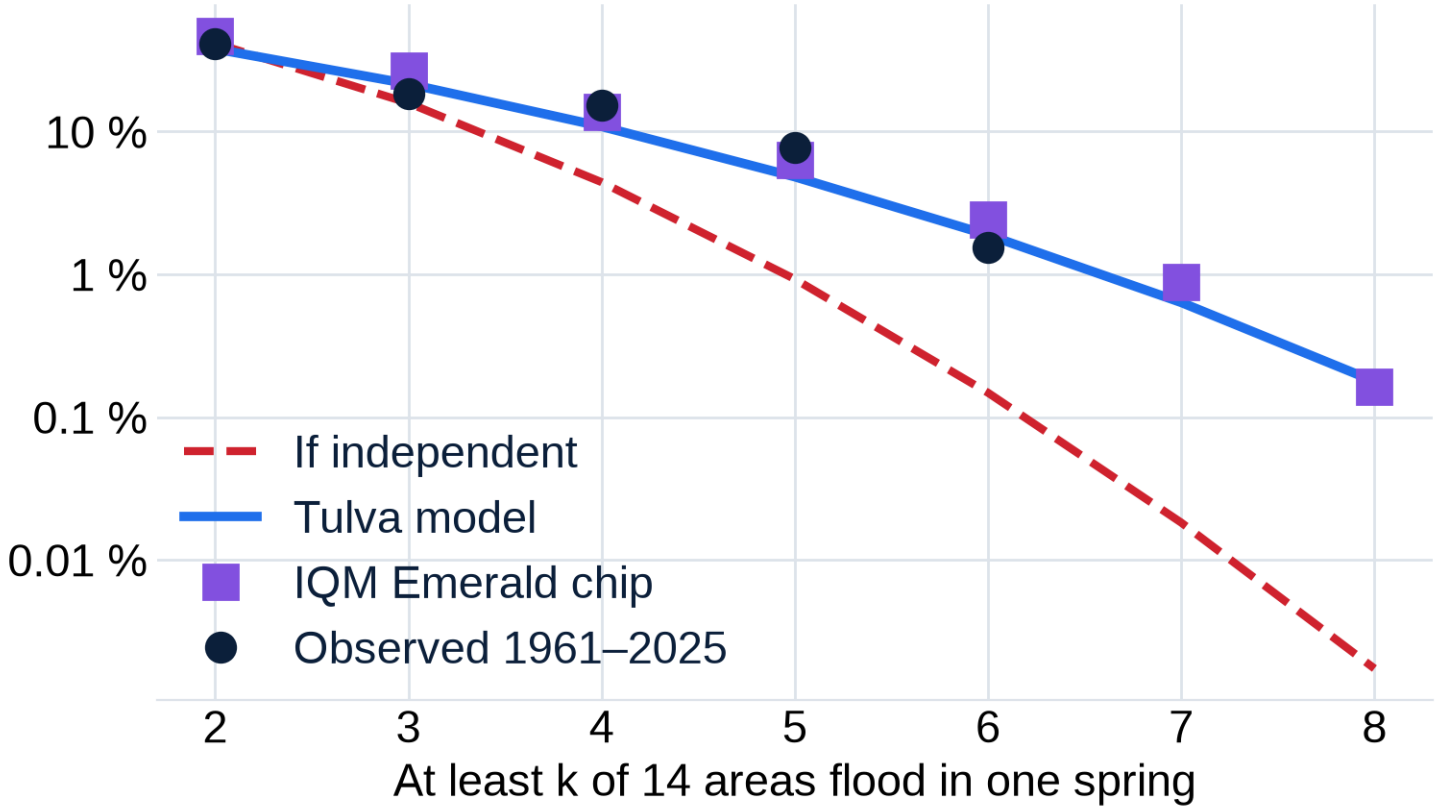
Source: SYKE 1961–2025, Tulva model (exact). Market price = 2.5–3.5× expected payout (AM Best, flood ILS)



Live app: joint-flood map → March snow → index price → quantum runs → 2005 replay

QUANTUM, IN PRACTICE

The full model fits on a chip today; amplitude estimation works on a 3-qubit piece of it.



At the 1-in-10 level the chip reproduces the joint tail.

ALL 14 AREAS ON IQM EMERALD

18 qubits

44 two-qubit gates

AMPLITUDE ESTIMATION

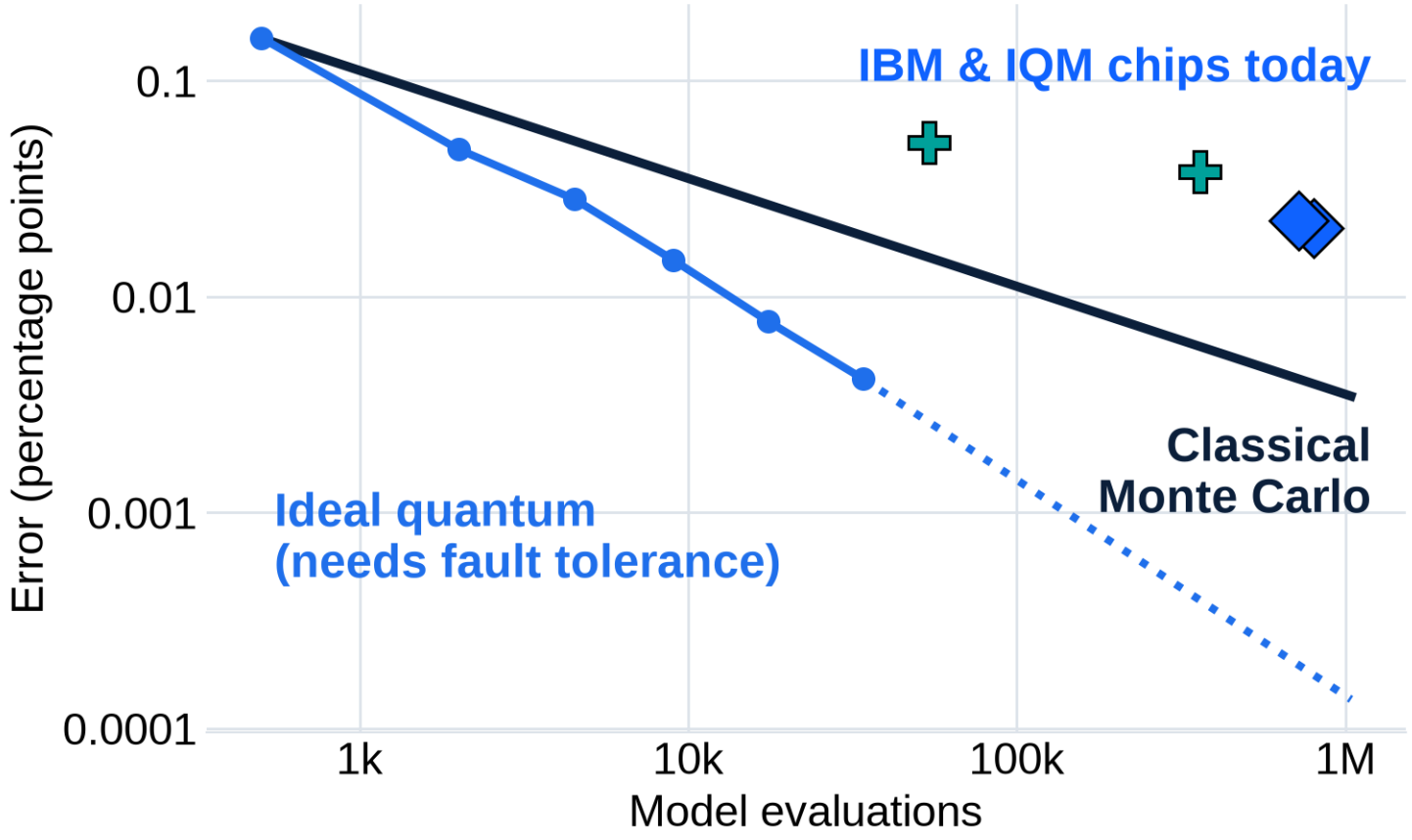
Kittilä and Ivalo both pass 1-in-50

Exact	**0.125 %**
IBM Marrakesh	**0.148 %**
IQM Garnet	**0.163 %**

Snow signal on the chip: snowy vs low March 3.1× (exact 2.7×)

IQM Emerald job 01a11cd4…, readout-corrected; IBM / IQM values calibrated (20 / 4 side-by-side copies); snow: Emerald, 10 copies

Classical Monte Carlo still wins today



Kittilä and Ivalo both pass 1-in-50: estimation error vs effort

Where quantum could pay off

Fault-tolerant hardware
~8k logical qubits needed

Expensive models
a hydrology model per scenario

More work per price
many perils, daily repricing

Error: QAE $\propto$ 1/N, Monte Carlo $\propto$ 1/√N

Source: Tulva simulations (qae-sim) + IBM Marrakesh / IQM Garnet runs; Chakrabarti et al., Quantum 5, 463 (2021)

BUSINESS

Who buys, why, and why now

WHO BUYS

Finnish non-life insurers

For the flood risk they keep. Four groups write 91 % of the market. Then reinsurers and ILS investors.

HOW WE EARN

Engine licence

Pricing and trigger engine, licensed per insurer or reinsurer. Fee on placed covers.

WHY NOW

30 Jan 2027

Finland gets a Solvency II flood capital factor: insurers need a joint-flood view now. The engine is quantum-ready as hardware matures.

SCALE

Finland → Nordics → Europe

Same engine, any snowmelt or river region. Gävle 2021: SEK 1.9bn claims. Storm Boris 2024: €2.15bn insured.

Ask **First step: a paid pilot with one Finnish insurer this autumn, before the 1 January reinsurance buy.**

Sources: Finance Finland 2024; Del. Reg. (EU) 2026/269; Finansinspektionen 2023 (Gävle); PERILS 2025 (Boris)

Tulva prices Finland's spring floods as a joint risk, and runs it on real quantum hardware.

BUILT THIS WEEKEND

SYKE data pipeline | Joint flood model | Index pricing | QAE on IBM & IQM

RESOURCES USED

SYKE open data | IBM Marrakesh | IQM Garnet & Emerald (Resonance) | Qrisp | Qiskit | AI coding agents

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack